

Amulet Motion Controller Fabrication Document

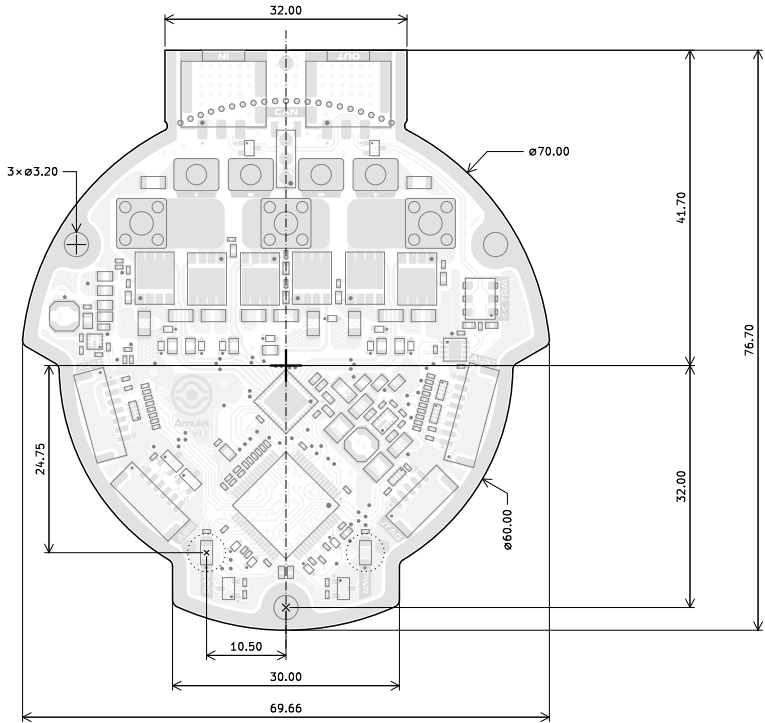
Top Fabrication (Scale 1:1)

Layer Stack Legend

|  | Material     | Layer         | Thickness     | Dielectric      | Type        | Gerber |
|--|--------------|---------------|---------------|-----------------|-------------|--------|
|  | F.Paste      |               |               |                 | Paste Mask  |        |
|  | F.Silkscreen |               |               |                 | Legend      | GBR    |
|  | F.Mask       |               | 0.02mm        | Solder Resist   | Solder Mask | GBR    |
|  | Copper       | L1 (Sig, PWR) | 0.07mm (2oz)  |                 | Signal      | GBR    |
|  | Prepreg      |               | 0.18mm        | FR4_7628        | Dielectric  |        |
|  | Copper       | L2 (GND)      | 0.035mm (1oz) |                 | Plane       | GBR    |
|  | Core         |               | 0.4mm         | FR4             | Dielectric  |        |
|  | Copper       | L3 (Sig, PWR) | 0.035mm (1oz) |                 | Signal      | GBR    |
|  | Prepreg      |               | 0.18mm        | FR4_7628        | Dielectric  |        |
|  | Copper       | L4 (Sig, PWR) | 0.035mm (1oz) |                 | Signal      | GBR    |
|  | Core         |               | 0.4mm         | FR4             | Dielectric  |        |
|  | Copper       | L5 (GND)      | 0.035mm (1oz) |                 | Plane       | GBR    |
|  | Prepreg      |               | 0.18mm        | FR4_7628        | Dielectric  |        |
|  | Copper       | L6 (Sig, PWR) | 0.07mm (2oz)  |                 | Signal      | GBR    |
|  | B.Mask       |               | 0.02mm        | Solder Resist   | Solder Mask | GBR    |
|  | B.Silkscreen |               |               | Direct Printing | Legend      | GBR    |
|  | B.Paste      |               |               |                 | Paste Mask  |        |

Total thickness: 1.66mm

Note: external layer thicknesses are specified after plating



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge\_Cuts.GBR" SUFFIX. DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DWG FOR REF ONLY.
- 2) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS. SELECTED HOLE LOCATIONS SHOWN ON THIS DWG FOR REF ONLY.
- 3) IMPEDANCE CONTROL REQUIRED. Microstrip 100-Ohm Differential (L1 ref. L2) 0.2032mm width, 0.28mm spacing
- 4) CONFIRM TRACE WIDTHS AND SPACINGS.
- 5) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

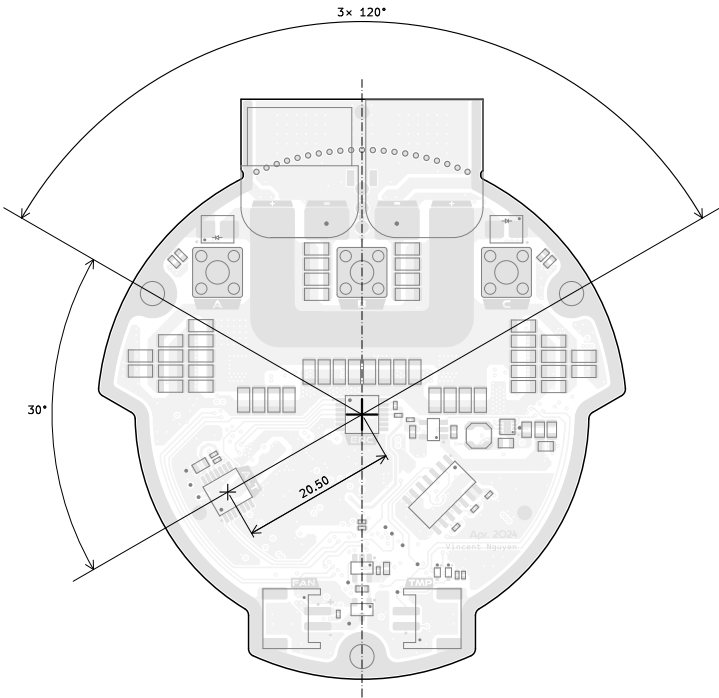
|                  |          |
|------------------|----------|
| TRACE WIDTH      | 0.20 mm  |
| TRACE TO TRACE   | 0.20 mm  |
| MIN. HOLE (PTH)  | 0.25 mm  |
| MIN. HOLE (NPTH) | 0.70 mm  |
| ANNULAR RING     | 0.15 mm  |
| COPPER TO HOLE   | 0.254 mm |
| COPPER TO EDGE   | 0.25 mm  |
| HOLE TO HOLE     | 0.254 mm |

All dimensions are in millimeters unless otherwise specified.

|                 |                             |                      |                                                                                       |               |          |           |
|-----------------|-----------------------------|----------------------|---------------------------------------------------------------------------------------|---------------|----------|-----------|
|                 | Comments:                   | Company:             |  |               | Variant: | Git Hash: |
|                 |                             | EPFL Xplore Research |                                                                                       |               | NO_XT60  | 1ffc255   |
|                 | Board Name:                 |                      |                                                                                       | Project Name: |          |           |
|                 | Amulet Motion Controller    |                      |                                                                                       | Chienpanzé    |          |           |
| Sheet Title:    | File Name:                  | Designer:            | Date:                                                                                 | Revision:     |          |           |
| Top Fabrication | amulet_controller.kicad_pcb | Vincent Nguyen       | 2024-04-13                                                                            | 1.1           |          |           |
| Sheet Path:     |                             | Reviewer:            | Size:                                                                                 | Sheet:        |          |           |
|                 |                             |                      | A4                                                                                    | 1 of 11       |          |           |

# Amulet Motion Controller Fabrication Document

Bottom Fabrication (Scale 1:1)



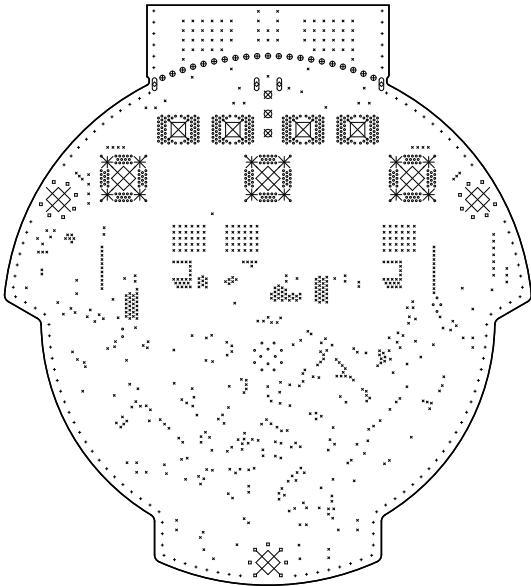
All dimensions are in millimeters unless otherwise specified.

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|--------------------|-----------------------------|--------------------------|---------------------------------------------------------------------------------------|-----------|---------------|-----------|
|                    | Comments:                   | Company:                 |  |           | Variant:      | Git Hash: |
|                    |                             | EPFL Xplore Research     |                                                                                       | NO_XT60   | 1ffc255       |           |
|                    |                             | Board Name:              |                                                                                       |           | Project Name: |           |
|                    |                             | Amulet Motion Controller |                                                                                       |           | Chienpanzé    |           |
| Sheet Title:       | File Name:                  | Designer:                | Date:                                                                                 | Revision: |               |           |
| Bottom Fabrication | amulet_controller.kicad_pcb | Vincent Nguyen           | 2024-04-13                                                                            | 1.1       |               |           |
| Sheet Path:        |                             | Reviewer:                | Size:                                                                                 | Sheet:    |               |           |
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# Amulet Motion Controller Fabrication Document

Drill Drawing (Top View) (Scale 1:1)

| Symbol | Count    | Hole Size | Plated   |
|--------|----------|-----------|----------|
| •      | 674      | 0.25mm    | Plated   |
| •      | 438      | 0.25mm    | Plated   |
| •      | 120      | 0.30mm    | Plated   |
| ▪      | 21       | 0.50mm    | Plated   |
| ◊      | 4 (slot) | 0.60mm    | Plated   |
| ⊠      | 3        | 1.00mm    | Plated   |
| ✱      | 12       | 1.50mm    | Plated   |
| ⊠      | 4        | 2.70mm    | Plated   |
| ⊠      | 6        | 3.20mm    | Plated   |
| •      | 21       | 0.70mm    | Unplated |

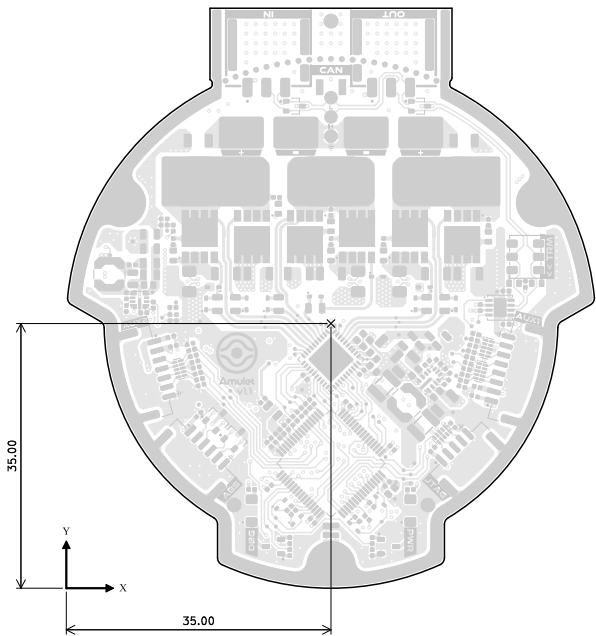


|  |                           |                                           |  |                                                                                       |           |                             |             |                      |  |
|--|---------------------------|-------------------------------------------|--|---------------------------------------------------------------------------------------|-----------|-----------------------------|-------------|----------------------|--|
|  | Comments:                 | Company:<br>EPFL Xplore Research          |  |  |           | Variant:<br>NO_XT60         |             | Git Hash:<br>1ffc255 |  |
|  |                           | Board Name:<br>Amulet Motion Controller   |  |                                                                                       |           | Project Name:<br>Chienpanzé |             |                      |  |
|  | Sheet Title:<br>Drill Map | File Name:<br>amulet_controller.kicad_pcb |  | Designer:<br>Vincent Nguyen                                                           |           | Date:<br>2024-04-13         |             | Revision:<br>1.1     |  |
|  | Sheet Path:               |                                           |  |                                                                                       | Reviewer: |                             | Size:<br>A4 | Sheet:<br>3 of 11    |  |

# Amulet Motion Controller Fabrication Document

| Ref. | Net | X | Y |
|------|-----|---|---|
|------|-----|---|---|

Top Test Points (Scale 1:1)



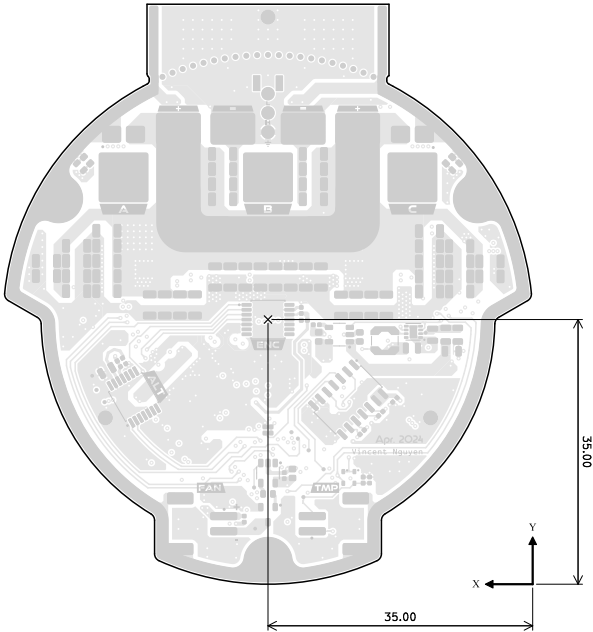
All dimensions are in millimeters unless otherwise specified.

|  |                                 |                                           |                             |                                                                                       |                             |  |                      |  |
|--|---------------------------------|-------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------|--|----------------------|--|
|  | Comments:                       | Company:<br>EPFL Xplore Research          |                             |  | Variant:<br>NO_XT60         |  | Git Hash:<br>1ffc255 |  |
|  |                                 | Board Name:<br>Amulet Motion Controller   |                             |                                                                                       | Project Name:<br>Chienpanzé |  |                      |  |
|  | Sheet Title:<br>Top Test Points | File Name:<br>amulet_controller.kicad_pcb | Designer:<br>Vincent Nguyen |                                                                                       | Date:<br>2024-04-13         |  | Revision:<br>1.1     |  |
|  | Sheet Path:                     |                                           | Reviewer:                   |                                                                                       | Size:<br>A4                 |  | Sheet:<br>4 of 11    |  |

# Amulet Motion Controller Fabrication Document

| Ref. | Net | X | Y |
|------|-----|---|---|
|------|-----|---|---|

Bottom Test Points (Scale 1:1)

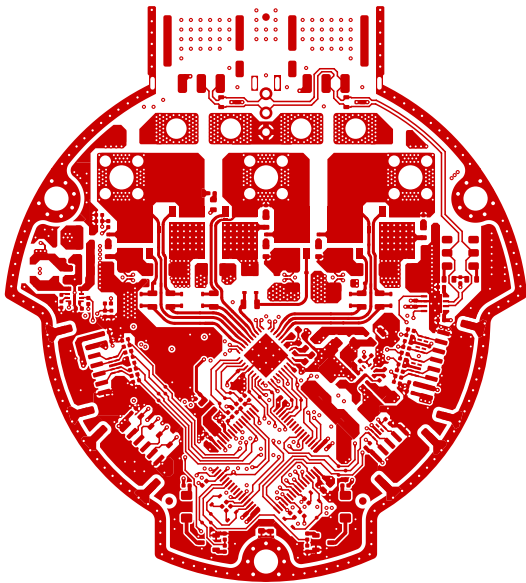


All dimensions are in millimeters unless otherwise specified.

|                    |                             |                          |                                                                                       |           |               |           |
|--------------------|-----------------------------|--------------------------|---------------------------------------------------------------------------------------|-----------|---------------|-----------|
|                    | Comments:                   | Company:                 |  |           | Variant:      | Git Hash: |
|                    |                             | EPFL Xplore Research     |                                                                                       |           | NO_XT60       | 1ffc255   |
|                    |                             | Board Name:              |                                                                                       |           | Project Name: |           |
|                    |                             | Amulet Motion Controller |                                                                                       |           | Chienpanzé    |           |
| Sheet Title:       | File Name:                  | Designer:                | Date:                                                                                 | Revision: |               |           |
| Bottom Test Points | amulet_controller.kicad_pcb | Vincent Nguyen           | 2024-04-13                                                                            | 1.1       |               |           |
| Sheet Path:        |                             | Reviewer:                | Size:                                                                                 | Sheet:    |               |           |
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# Amulet Motion Controller Fabrication Document

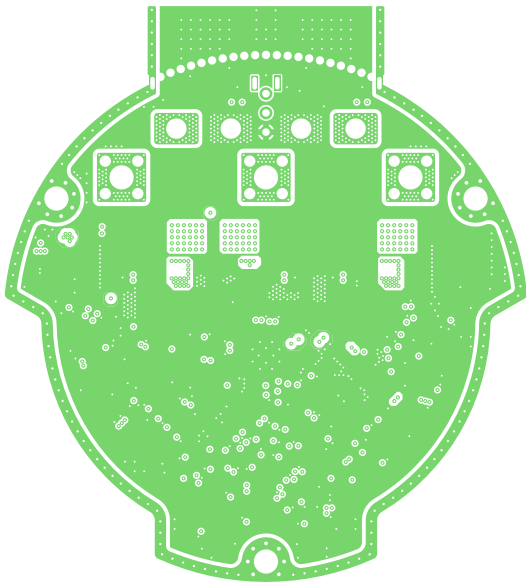
L1 (SIG, PWR)



|  |                      |                                           |                             |                                                                                       |                             |                     |                      |                   |
|--|----------------------|-------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------|---------------------|----------------------|-------------------|
|  | Comments:            | Company:<br>EPFL Xplore Research          |                             |  | Variant:<br>NO_XT60         |                     | Git Hash:<br>1ffc255 |                   |
|  |                      | Board Name:<br>Amulet Motion Controller   |                             |                                                                                       | Project Name:<br>Chienpanzé |                     |                      |                   |
|  | Sheet Title:<br>F.Cu | File Name:<br>amulet_controller.kicad_pcb | Designer:<br>Vincent Nguyen |                                                                                       |                             | Date:<br>2024-04-13 |                      | Revision:<br>1.1  |
|  | Sheet Path:          |                                           |                             | Reviewer:                                                                             |                             |                     | Size:<br>A4          | Sheet:<br>6 of 11 |

# Amulet Motion Controller Fabrication Document

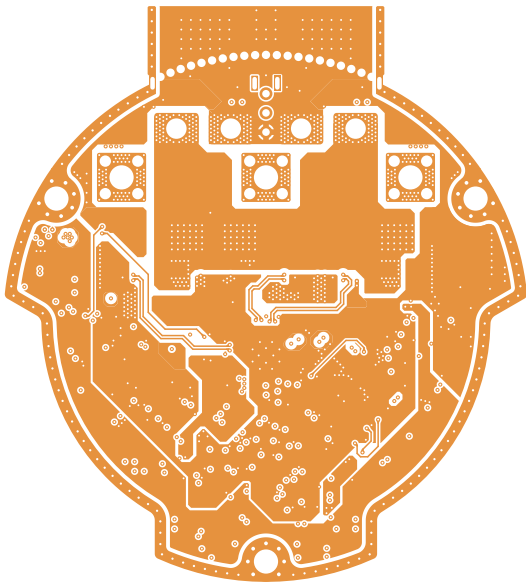
L2 (GND)



|              |           |                             |                          |                |                                                                                       |            |               |           |  |  |
|--------------|-----------|-----------------------------|--------------------------|----------------|---------------------------------------------------------------------------------------|------------|---------------|-----------|--|--|
|              | Comments: |                             | Company:                 |                |  | Variant:   |               | Git Hash: |  |  |
|              |           |                             | EPFL Xplore Research     |                |                                                                                       | NO_XT60    |               | 1ffc255   |  |  |
|              |           |                             | Board Name:              |                |                                                                                       |            | Project Name: |           |  |  |
|              |           |                             | Amulet Motion Controller |                |                                                                                       |            | Chienpanzé    |           |  |  |
| Sheet Title: |           | File Name:                  |                          | Designer:      |                                                                                       | Date:      |               | Revision: |  |  |
| In1.Cu       |           | amulet_controller.kicad_pcb |                          | Vincent Nguyen |                                                                                       | 2024-04-13 |               | 1.1       |  |  |
| Sheet Path:  |           |                             |                          | Reviewer:      |                                                                                       | Size:      |               | Sheet:    |  |  |
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# Amulet Motion Controller Fabrication Document

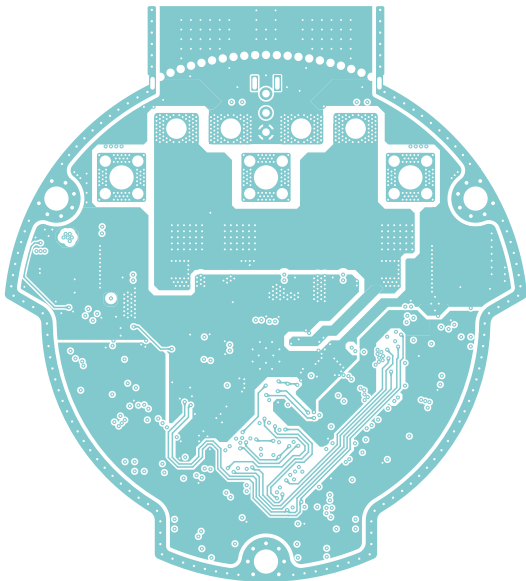
L3 (SIG, PWR)



|  |                        |                                           |                             |                                                                                       |                             |                     |                      |                   |
|--|------------------------|-------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------|---------------------|----------------------|-------------------|
|  | Comments:              | Company:<br>EPFL Xplore Research          |                             |  | Variant:<br>NO_XT60         |                     | Git Hash:<br>1ffc255 |                   |
|  |                        | Board Name:<br>Amulet Motion Controller   |                             |                                                                                       | Project Name:<br>Chienpanzé |                     |                      |                   |
|  | Sheet Title:<br>In2.Cu | File Name:<br>amulet_controller.kicad_pcb | Designer:<br>Vincent Nguyen |                                                                                       |                             | Date:<br>2024-04-13 |                      | Revision:<br>1.1  |
|  | Sheet Path:            |                                           |                             | Reviewer:                                                                             |                             |                     | Size:<br>A4          | Sheet:<br>8 of 11 |

# Amulet Motion Controller Fabrication Document

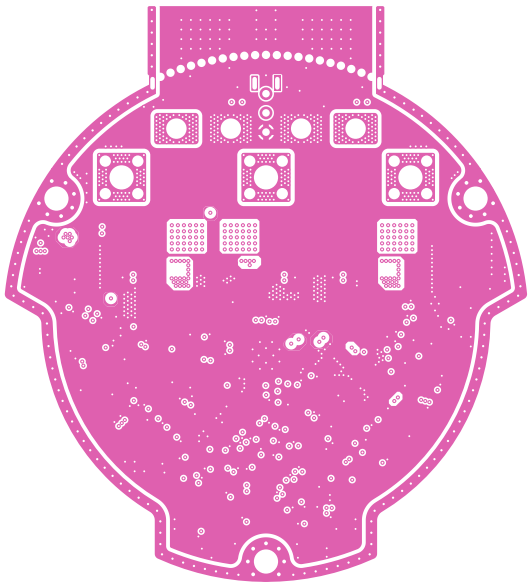
L4 (SIG, PWR)



|              |                             |                      |                                                                                       |           |          |                             |
|--------------|-----------------------------|----------------------|---------------------------------------------------------------------------------------|-----------|----------|-----------------------------|
|              | Comments:                   | Company:             |  |           | Variant: | Git Hash:                   |
|              |                             | EPFL Xplore Research |                                                                                       |           | NO_XT60  | 1ffc255                     |
|              |                             | Board Name:          | Amulet Motion Controller                                                              |           |          | Project Name:<br>Chienpanzé |
| Sheet Title: | File Name:                  | Designer:            | Date:                                                                                 | Revision: |          |                             |
| In3.Cu       | amulet_controller.kicad_pcb | Vincent Nguyen       | 2024-04-13                                                                            | 1.1       |          |                             |
| Sheet Path:  |                             | Reviewer:            | Size:                                                                                 | Sheet:    |          |                             |
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# Amulet Motion Controller Fabrication Document

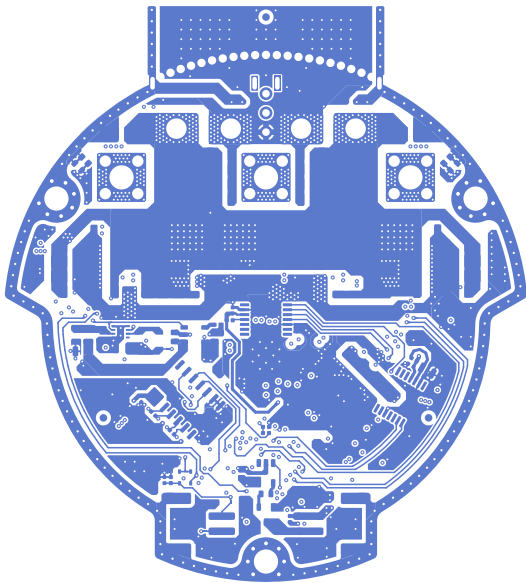
L5 (GND)



|  |                        |                                           |                             |                                                                                       |                             |                      |
|--|------------------------|-------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------|----------------------|
|  | Comments:              | Company:<br>EPFL Xplore Research          |                             |  | Variant:<br>NO_XT60         | Git Hash:<br>1ffc255 |
|  |                        | Board Name:<br>Amulet Motion Controller   |                             |                                                                                       | Project Name:<br>Chienpanzé |                      |
|  | Sheet Title:<br>In4.Cu | File Name:<br>amulet_controller.kicad_pcb | Designer:<br>Vincent Nguyen |                                                                                       | Date:<br>2024-04-13         | Revision:<br>1.1     |
|  | Sheet Path:            |                                           | Reviewer:                   |                                                                                       | Size:<br>A4                 | Sheet:<br>10 of 11   |

# Amulet Motion Controller Fabrication Document

L6 (SIG, PWR)



|  |                      |                                           |                             |                                                                                       |                             |                     |                      |                    |
|--|----------------------|-------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------|---------------------|----------------------|--------------------|
|  | Comments:            | Company:<br>EPFL Xplore Research          |                             |  | Variant:<br>NO_XT60         |                     | Git Hash:<br>1ffc255 |                    |
|  |                      | Board Name:<br>Amulet Motion Controller   |                             |                                                                                       | Project Name:<br>Chienpanzé |                     |                      |                    |
|  | Sheet Title:<br>B.Cu | File Name:<br>amulet_controller.kicad_pcb | Designer:<br>Vincent Nguyen |                                                                                       |                             | Date:<br>2024-04-13 |                      | Revision:<br>1.1   |
|  | Sheet Path:          |                                           |                             | Reviewer:                                                                             |                             |                     | Size:<br>A4          | Sheet:<br>11 of 11 |